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Organizational Buying Behavior: Toward an Integrative Framework

Wesley J. Johnston
GEORGIA STATE UNIVERSITY
Jeffrey E. Lewin
GEORGIA STATE UNIVERSITY

This article analyzes and summarizes the 25 years of research on organizational buying behavior that followed the seminal works of Robinson, Faris, and Wind (1967), Webster and Wind (1972), and Sheth (1973). Based on a review of 165 articles, the authors present an integrated model of organizational buying behavior that both combines the propositions of the original three works and incorporates several new constructs that have emerged over the past 25 years. In addition, the authors identify some of the important findings associated with the past research in organizational buying behavior. Each article citation in the Reference Appendix includes a list of the organizational buying behavior constructs examined within the article. J BUSN RES 1996. 35.1–15

In order to succeed in business-to-business markets, selling firms must possess an understanding of customer firms' buying behavior. However, such an understanding may be difficult to achieve, because organizational buying behavior is often a multiphase, multiperson, multidepartmental, and multiobjective process. This dynamic and intricate process frequently presents sellers with a complex set of issues and situational factors that directly or indirectly influence buying firm behavior.

In the late 1960s, interest in studying and understanding the organizational buying process began to increase dramatically. In 1967, Robinson, Faris, and Wind published their seminal book *Industrial Buying and Creative Marketing*, which included a model of "the industrial buying process" and the "buygrid framework." A few years later (1972) Webster and Wind presented their "general model for understanding organizational buying behavior" and in 1973 Sheth published his "model of industrial buyer behavior." Together, these three works laid the conceptual foundation for the study of organizational buying behavior. Over the past 25 years, hundreds of conceptual and empirical articles have been published that either extend or test (part or all) of the models proposed by these authors.

However, with very few exceptions (Bonoma, Zaltman, and Johnston, 1977; Moriarty and Galper, 1978; Ward and Webster, 1991), little has been done to organize and summarize the findings and knowledge gained from all this research. Our purpose is to examine and integrate the stream of research on organizational buying behavior that has emerged from these three early works. Our search is limited to articles that have been published in the following marketing journals: *Journal of Marketing*, *Journal of Marketing Research*, *Journal of Business Research*, *Industrial Marketing Management*, *Journal of Business & Industrial Marketing*, and *Journal of Business-to-Business Marketing*. Based on the results of our examination, we present an integrated model of organizational buying behavior that combines the constructs and variables included in the three original models. In addition, we discuss four new constructs that have emerged as important antecedents or consequences of organizational buying behavior.

Our discussion is divided into five parts. First, we briefly summarize and compare the original three models in order to form a composite model. Second, based on our analysis we extend the original models by adding four additional constructs. Third, we analyze the stream of research that conceptually and empirically scrutinized the original models. We then offer our view of what has been learned about organizational buying behavior over the past 25 years. Finally, we suggest some directions for future research.

Organizational Buying: The Three Original Models

Central to the models proposed by Robinson, Faris, and Wind; Webster and Wind; and Sheth is the concept that organizational buying behavior is a process. Within their "buygrid frame-

Address correspondence to: Wesley J. Johnston, Department of Marketing, Georgia State University, Atlanta, GA 30303.

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work," Robinson, Faris, and Wind introduced their "buyphases." These phases (or stages) represent the sequence of activities often performed in an organizational buying situation. Briefly, these activities are: (1) recognition of need and a general solution, (2) determination of characteristics and quantity, (3) description of characteristics and quantity, (4) search for potential sources, (5) acquire and analyze proposals, (6) evaluate proposals and select supplier(s), (7) select an order routine, and (8) performance feedback and evaluation. Similarly, both the Webster and Wind and Sheth models presented organizational buying behavior as a process composed of a sequence of phases or stages. Although each of these two subsequent models used fewer than eight stages (five and four, respectively), the nature and sequence of events to be performed remained quite similar.

In addition to agreeing that organizational buying behavior is best described as a process, these three original models share several other important constructs. First, each model contains environmental influences such as physical, political, economic, suppliers, competitors, technological, legal, cultural, and global (in the Sheth model these are called situational influences). Second, each model contains organizational influences, such as size, structure, orientation, technology, rewards, tasks, and goals. Finally, all three models contain individual participants' (buyers') characteristics that include education, motivation, perceptions, personality, risk reduction, and experience.

The Robinson, Faris, and Wind model and the Sheth model have two additional constructs in common. The first of these, purchase (product) characteristics, includes variables such as buy task, product type, perceived risk, prior experience, product complexity, and time pressure. The second, seller characteristics, or the criteria by which potential vendors are evaluated, include price, the ability to meet specifications, product quality, delivery time, and after-sale service.

A sixth construct, group characteristics, is introduced in Webster and Wind's general model for understanding organizational buying behavior. These characteristics include size, structure, authority, membership, experiences, expectations, leadership, objectives, and backgrounds. Webster and Wind argued that every buying center is unique, and that it is the special combination of organizational and group characteristics that contributes to this uniqueness.

Two final constructs are introduced in Sheth's model of industrial buyer behavior. The first is informational characteristics or the source and type of information each decision-maker is exposed to and his/her participation in the active search for this information. Suggested sources and types of information include salespeople, conferences and trade shows, word-of-mouth, trade news, direct mail, and advertising. The second construct introduced by Sheth is conflict negotiation characteristics. Sheth posited that decision-makers employ a variety of methods to resolve the inevitable conflicts associated with joint decision-making. Adopting the March and Simon (1958) typology, Sheth argued that these resolution methods include problem solving and persuasion (which are useful and rational), as well as bargaining and politicking (which are nonrational and inefficient).

These nine constructs (environmental, organizational, group, participant, purchase, seller, conflict/negotiation, informational and process or stages) broadly represent the variables found in the original three models that combine to affect organizational buying behavior. In retrospect, perhaps it's not surprising to find that these three original models are so fundamentally similar. After all, each model is describing the same basic organizational buying process, and each was developed at relatively the same time. Further, after 25 years of empirical testing, it appears that these models were correct in proposing that environmental, organizational, group, participant, purchase, seller, informational, and conflict/negotiation characteristics, as well as the stages in the buying process significantly affect organizational buying behavior.

Extending the Original Models

The macromodels proposed by Robinson, Faris, and Wind (1967), Webster and Wind (1972), and Sheth (1973) provided the general categories of constructs expected to influence organizational buying behavior. However, models constructed during the early stages of theory development often fail to capture all of the concepts, variables, and relationships needed to consistently predict complex behavioral outcomes. In addition, it is not unusual for constructs originally modeled as unidimensional to be recognized, under closer scrutiny, as being multidimensional. Therefore, as a theory begins to mature, both conceptually and empirically, additional constructs and relationships frequently emerge as important predictors of behavior. In the study of organizational buying behavior four such constructs have emerged. The first two, role stress and decision rules, operate at the intrafirm level of organizational buying behavior. The second two, buyer–seller relationships and communication networks operate at the interfirm level.

Intrafirm

THE RELATIONSHIPS AMONG THE ORIGINAL NINE CONSTRUCTS. Figure 1 incorporates the nine original constructs (previously discussed) as well as two additional constructs – decision rules and role stress – which we believe, based on our review, needed to be included. These two additional constructs will be discussed shortly.

We determined the nature (direction) of the relationships between the constructs depicted in Figure 1 in the following manner. First, the theoretical relationships proposed in the original three models, as well as sound theoretical relationships proposed in subsequent conceptual articles were incorporated into Figure 1. And, second, if in the absence of or contrary to theory, *significant* empirical support was found for certain additional relationships, these other relationships were also incorporated into our integrated model.

As shown in Figure 1, environmental, organizational, pur-

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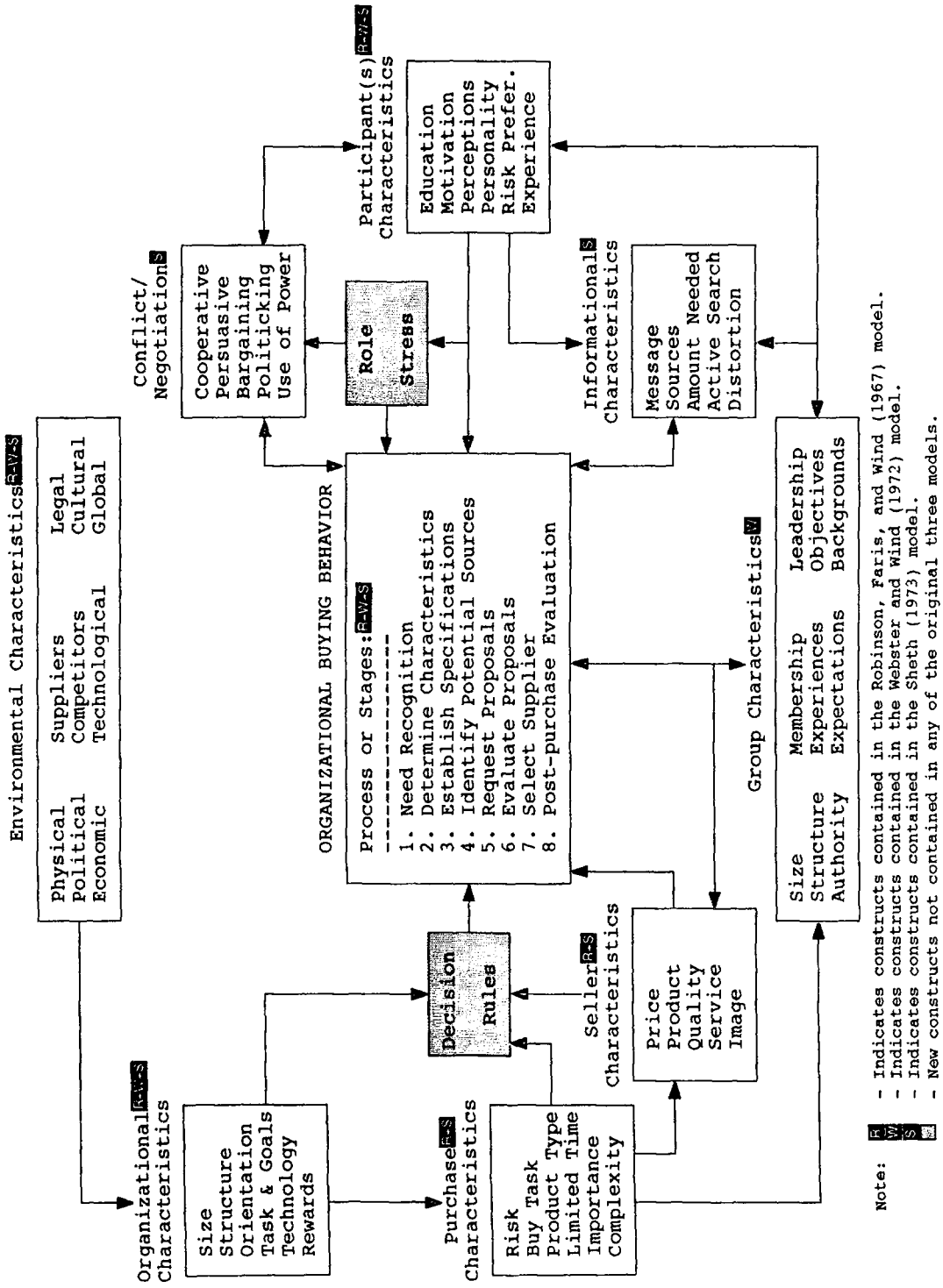
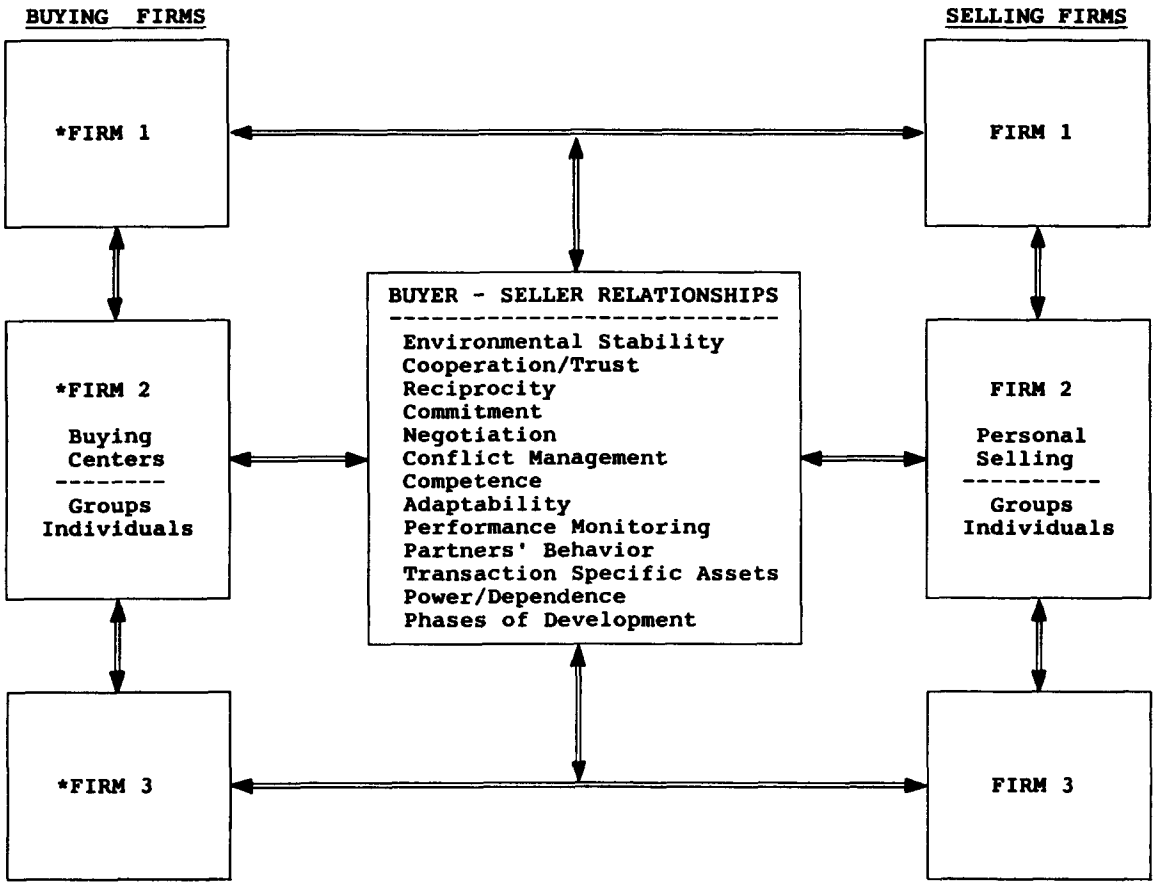


Figure 1. An integrated model of organizational buying behavior.



* We assume that the integrated model of organizational buying behavior, as proposed in Figure 1, applies to each of the three "Buying Firms" in Figure 2.

** ⟷ Select communication pathways between firms.

Figure 2. Buyer–Seller relationships and communication networks.

chase, group, participants, conflict/negotiation, informational, and seller characteristics are expected to influence, either directly and/or indirectly, the organizational buying process. Additionally, the relationship among three constructs—group, informational, and conflict/negotiation—and the organizational buying process is expected to be bidirectional. That is, like the other constructs, group, informational, and conflict/negotiation characteristics are expected to influence the organizational buying process. However, the organizational buying process is also expected to influence the characteristics of the buying group (e.g., size and membership), the amount and types of information used (e.g., personal/impersonal, commercial/noncommercial), the nature of the conflict experienced (e.g., high/low, latent/manifest) and the negotiation strategies used (e.g., cooperative/bargaining) throughout the different stages of the organizational buying process.

DECISION RULES AND ROLE STRESS. As Figure 1 shows, we have incorporated two additional constructs into our integrated model, decision rules and role stress. As shown later, these two constructs have been used in a number of conceptual and empirical articles that proposed and found support for their effect on organizational buying behavior.

The first construct, decision rules, is influenced by environmental, organizational, purchase, and seller characteristics, and is expected to vary across the stages of the organizational buying process. In some organizations these are formalized rules and procedures, which may include formulas for selecting suppliers. In other organizations these are informal rules based on buyers' experience and rules-of-thumb (Vyas and Woodside, 1984; extended in Woodside, 1988).

The second, additional construct, role stress, is influenced by environmental, organizational, purchase, group, and participant(s) characteristics. In the study of organizational buying behavior, role stress is frequently operationalized as role ambiguity and/or role conflict. Role conflict is the degree of incongruity or incompatibility among purchase expectations. Role ambiguity is the degree to which clear information is lacking about: (1) the expectations associated with a purchase, (2) the methods for fulfilling known purchase expectations, and/or

(3) the consequences of role performance. For example, a directive from management ostensibly mandates both a decrease in the cost of certain key components and a concurrent increase in the quality of these same components. How can buying center members reconcile these seemingly conflicting expectations, and what will the consequences be if they fail to meet management's expectations?

Interfirm

Thus far we have been discussing organizational buying behavior at the intrafirm level. Based on the work we reviewed, it seems clear that all but the smallest of firms purchase many of their goods and services using the process known as organizational buying behavior. Accordingly, our integrated model could be used to describe the activities taking place within most buying firms. However, organizational buying behavior is often conceptualized in the research literature as a dyadic process requiring significant interfirm interaction. On one side of the dyad are the activities that take place within the buying firm. On the other side of the dyad are the activities that take place within the selling firm. Although it is beyond the scope of this article to describe selling firm activities, these activities have been studied and shown to exist. In fact, some authors have used the term "selling center" to describe these activities and the group of people who perform them (Woodside et al., 1977; Hutt, Johnston, and Ronchetto, 1985; Spekman and Johnston, 1986; Puri and Korgaonkar, 1991; Puri, 1992). Of the 165 articles reviewed, 49 discussed and/or examined organizational buying behavior in a dyadic or network-of-firms perspective.

The model presented in Figure 2, buyer–seller relationships and communication networks, provides a framework for discussing the interfirm aspects of organizational buying behavior, as well as the final two important constructs previously used to investigate and explain organizational buying behavior, buyer–seller relationships and communication networks.

BUYER-SELLER RELATIONSHIPS. In Figure 2, we illustrate both the dyadic and network perspectives of organizational buying behavior. We assume that our integrated model of organization buying behavior, as presented in Figure 1, applies to each of the three buying firms. That is, within each of the three buying firms depicted in Figure 2, the process of organization buying behavior, as presented in Figure 1, is taking place. In addition, we assume that each of the three selling firms has a selling center. We arbitrarily select buying firm 2 and selling firm 2 to focus our attention on the dyadic issues of the buyer–seller relationships construct. As can be seen in Figure 2, organizational buying behavior, which takes place within the buying firm, affects and is affected by buyer–seller relationships. Implicit in this proposition is the view that all of those factors that affect organizational buying behavior combine to affect a firm's relationship with any given supplier. Within the frame of our review, the buyer–seller relationship construct ranked eighth in frequency of use. The variables most often used to examine this construct include power/dependence, behavior/performance monitoring, cooperation/trust, adaptability, and commitment.

COMMUNICATION NETWORKS. Communication networks, the last construct we include in our analysis of organizational buying behavior, is unique. Whereas all of the other constructs are discussed as existing in either an intrafirm or interfirm context, communications networks exist in both. That is, members of the buying firm involved in the purchase process communicate with other buying center members as well as with co-workers outside of the buying center (intrafirm). In addition, members of a buying center communicate with one or more selling center members during the course of the purchase process (interfirm). Finally, members from different buying firms frequently communicate with one another as well as with members from different selling firms through participation in trade shows, conferences, exhibitions, and professional associations (interfirm).

The model presented in Figure 2 partially reflects the extensity of the interfirm aspect of organizational communication networks. That is, the bidirectional pathways depicted in Figure 2 indicate that buying firms communicate with other buying firms, as well as with various selling firms. Similarly, selling firms communicate with one other, as well as with various buying firms. Also, the communication networks that exist between buying and selling firms have both a direct and an indirect effect on specific buyer–seller relationships.

Due to the nature of our literature review, we examined only 13 articles that used communication networks in their study of organizational buying behavior. However, the research in organizational buying behavior examining the communication network construct is only part of the total research in this area. Communication network research spans several disciplines and research perspectives. Had we expanded our review to include additional research areas in marketing (i.e., channels of distribution or personal selling) or areas outside of marketing (i.e., organizational behavior, management, or sociology of organizations) we are confident that the number of reviewed articles examining the communication network construct would have increased dramatically.

Twenty-five Years of Empirical and Conceptual Scrutiny

As discussed in the introduction, our literature search was limited to the following six marketing journals: *Journal of Marketing, Journal of Marketing Research, Journal of Business Research, Industrial Marketing Management, Journal of Business & Industrial Marketing*, and *Journal of Business-to-Business Marketing*. We selected these journals for the following reasons:

1. The *Journal of Marketing* and *Journal of Marketing Research* are the leading academic journals and are broad-based in their editorial content.
2. The *Journal of Business & Industrial Marketing, Journal of*

Business-to-Business Marketing, and *Industrial Marketing Management* are the three leading journals specializing in organizational buying behavior.

3. The *Journal of Business Research* has shown a special interest in organizational buying behavior, publishing numerous individual articles and two special issues on this topic.

We should also point out that only articles dealing directly with organizational buying behavior were examined. Articles focusing on topics such as personal selling, channels of distribution, or advertising and promotion were excluded from our review.

Conceptual and Empirical Articles

Table 1 presents the 13 constructs that permeate the organizational buying behavior literature we reviewed. The table is arranged so that the constructs are ranked according to how frequently they have been used to explain organizational buying behavior. The tally column represents the total number of articles, either conceptual or empirical, that used a particular construct. In other words, any time one or more of these 13 constructs was examined, in either an empirical or conceptual article, each construct used was scored in our categorization. Finally, the column labeled "Reference Key Word" has special meaning. We have included a listing of the constructs used in the reviewed articles at the end of each citation in the Reference Appendix. Each construct listed at the end of a citation can be identified by its "key word" designation. For example, the first citation in the Reference Appendix is:

> Abratt, Russell, Industrial Buying in High-Tech Markets. *Industrial Marketing Management* 15 (1986): 293–298. [Process, Group, Information, Seller]

Thus, the constructs process or stages, group characteristics, informational characteristics, and seller characteristics, were examined in this article.

In addition to the general information presented in Table 1, some specific findings on certain individual constructs are noteworthy. First, with regard to the most frequently used construct, purchase characteristics, six variables have been used repeatedly as predictors of organizational buying behavior. In their order of frequency these variables are product type (capital equipment, component parts, materials, supplies, and/or services), buy task (new task, modified rebuy, and/or straight rebuy), purchase risk/uncertainty, purchase complexity, time pressure, and purchase importance. Second, in many studies it is difficult to separate the group and participant(s) constructs because individual participant characteristics are aggregated to form the buying center's group characteristics. However, with this in mind, the following group participant(s) variables have frequently been used as predictors of organizational buying behavior: group membership, group size, group structure, influence, experience, and demographics. Third, in most of the studies that used the seller characteristics construct, seller characteristics were conceptualized as supplier selection criteria.

Table 1. Determinants of Organizational Buying Behavior

Rank	Construct	Tally	Reference Key Word
1	Purchase characteristics	67	Purchase
2	Organizational characteristics	65	Organization
3	Group characteristics	65	Group
4	Participant(s) characteristics	55	Participant
5	Process or stages	48	Process
6	Seller characteristics	48	Seller
7	Informational characteristics	44	Information
8	Buyer-Seller relationships	36	Relationship
9	Environmental characteristics	31	Environment
10	Conflict/negotiation tactics	24	Conflict/negotiation
11	Communication networks	13	Network
12	Decision rules	12	Rules
13	Role stress	6	Stress

The most frequently used selection criterion variables include product quality, price, delivery, service (i.e., technical support, flexibility), past performance, and vendor reputation and image. Finally, when the informational characteristics construct was used, the most common variable sets were sources and types of information, followed by levels of information needed, and message content.

Empirical Articles Only

In the next step of our analysis we separated the 44 empirical from the 121 conceptual articles. We then examined the following aspects of the empirical articles: (1) the constructs used in each article, (2) the variables used to represent each construct, (3) the relationship among the variables (dependent, independent, correlated), and (4) the statistical significance of each relationship.

Figure 3 presents the findings of this analysis. In Figure 3 the matrix columns provide information on each construct when it was used as a dependent variable, and the rows provide information on each construct when it was used as an independent variable. For example, $cell_{1,2}$ indicates that purchase characteristics was used only once as an independent variable to predict the dependent variable organizational characteristics, and the relationship was statistically significant at $p \leq .10$. Similarly, $cell_{2,1}$ indicates that organizational characteristics was used only once as an independent variable to predict purchase characteristics, however the relationship was not statistically significant (*NS*). The diagonal cells provide information on studies that used one or more aspects of a construct to predict other aspects of the same construct. For example, levels and types of conflict are often used as predictors of negotiation tactics. In our review, five studies found one or more aspects of conflict/negotiation to be statistically significant predictors of other aspects of conflict/negotiation, ($cell_{10,10}$). Finally, if a study simply examined the correlation between constructs, each construct was considered to be an independent variable.

Based on the information provided in Figure 3, it seems clear

		Used As Dependent Variables												
		Purchase Charac.	Organiz. Charac.	Group Charac.	Partic. Charac.	Process/ Stages	Seller Charac.	Inform. Charac.	B-S Relations	Environ. Charac.	Conflict/ Negotiat.	Commun. Networks	Decision Rules	Role Stress
Used As Independent Variables	Purchase Charac.	1 signif. 0 n.s.	1 signif. 0 n.s.	17 signif. 0 n.s.	3 signif. 0 n.s.	3 signif. 2 n.s.	9 signif. 1 n.s.	7 signif. 0 n.s.	4 signif. 0 n.s.		6 signif. 0 n.s.	2 signif. 0 n.s.	3 signif. 0 n.s.	
	Organiz. Charac.	0 signif. 1 n.s.	2 signif. 0 n.s.	7 signif. 1 n.s.	5 signif. 0 n.s.	2 signif. 1 n.s.	6 signif. 0 n.s.	5 signif. 1 n.s.	1 signif. 1 n.s.		5 signif. 0 n.s.	2 signif. 0 n.s.	1 signif. 0 n.s.	
	Group Charac.			4 signif. 0 n.s.	3 signif. 0 n.s.	2 signif. 0 n.s.	4 signif. 1 n.s.				1 signif. 0 n.s.			
	Partic. Charac.	0 signif. 1 n.s.		5 signif. 0 n.s.	5 signif. 0 n.s.	4 signif. 0 n.s.	5 signif. 1 n.s.	6 signif. 2 n.s.	1 signif. 0 n.s.		2 signif. 1 n.s.	1 signif. 0 n.s.	1 signif. 0 n.s.	
	Process/ Stages			8 signif. 1 n.s.	3 signif. 0 n.s.	1 signif. 0 n.s.		1 signif. 0 n.s.					1 signif. 0 n.s.	
	Seller Charac.				1 signif. 1 n.s.	7 signif. 0 n.s.	0 signif. 2 n.s.	2 signif. 0 n.s.	2 signif. 0 n.s.		3 signif. 0 n.s.			
	Inform. Charac.	1 signif. 0 n.s.	1 signif. 0 n.s.	1 signif. 0 n.s.	2 signif. 0 n.s.	5 signif. 0 n.s.					2 signif. 0 n.s.		1 signif. 0 n.s.	
	B-S Relations		0 signif. 1 n.s.	1 signif. 1 n.s.	1 signif. 0 n.s.	2 signif. 0 n.s.	2 signif. 0 n.s.	1 signif. 0 n.s.	2 signif. 1 n.s.		2 signif. 1 n.s.		2 signif. 0 n.s.	
	Environ. Charac.			3 signif. 0 n.s.		0 signif. 1 n.s.	2 signif. 0 n.s.	2 signif. 0 n.s.	3 signif. 1 n.s.		1 signif. 0 n.s.		2 signif. 0 n.s.	
	Conflict/ Negotiat.				7 signif. 0 n.s.				1 signif. 0 n.s.		5 signif. 0 n.s.			
	Commun. Network			2 signif. 1 n.s.			1 signif. 0 n.s.				1 signif. 0 n.s.	0 signif. 1 n.s.		
	Decision Rules			2 signif. 0 n.s.	1 signif. 0 n.s.	5 signif. 0 n.s.	1 signif. 0 n.s.		1 signif. 0 n.s.				1 signif. 0 n.s.	
	Role Stress			1 signif. 0 n.s.	1 signif. 0 n.s.				1 signif. 0 n.s.		1 signif. 0 n.s.			

Figure 3. Constructs used in empirical research on organizational buying behavior. Note: *All* statistical significance (signif.) is $p \leqslant .10$; however, *with only a few exceptions* statistical significance is $p \leqslant .05$.

that some constructs have been used primarily as independent variables (e.g., purchase, organizational, and environmental characteristics), whereas other constructs have been used primarily as dependent variables (e.g., group characteristics and process/stages). Still other constructs have been used with equal frequency as both dependent and independent variables (e.g., buyer-seller relationships and communication networks).

After 25 Years, What Have We Learned?

In examining organizational buying behavior, our review has covered a little over 25 years of research. We have focused on the conceptual and empirical research stream that was guided and influenced by Robinson, Faris, and Wind (1967), Webster and Wind (1972), and Sheth (1973). Although we limited our review to only six high-quality marketing journals, we examined 165 empirical and conceptual articles. Drawing on the information contained in these reviewed articles, we offer the following macro-view of how the 13 constructs previously discussed in this article influence and affect organizational buying behavior.

Dynamics of Organizational Buying Behavior

Figure 4A provides a framework for discussing our macro-view of organizational buying behavior. As Robinson, Faris, and Wind (1967) speculated in their concept of buy task, much of the variation in organizational buying behavior appears to be related to the levels of risk associated with a given purchase situation. Further, purchase risk, in most cases, is a function of: (1) the importance of a particular purchase, (2) the complexity associated with the purchase, (3) the uncertainty of the purchase outcome, and/or (4) the need to reach a decision quickly (time pressure).

Any number or variety of variables can contribute to the level of risk associated with a purchase situation. However, most of these antecedent variables fall within the constructs of environmental, organizational, or purchase characteristics. Examples of these variables include the level of environmental uncertainty (high versus low), the characteristics of the *buyer* firm (size, technical competence), the type of product to be purchased (capital equipment versus supplies), and the nature of the buy task (new task versus rebuy).

In addition, two aspects of both buyer–seller relationships and communication networks have the ability to influence the amount of risk associated with a particular purchase situation. The first is simply whether or not a prior relationship or channel of communication exists between the buying firm and one or more potential suppliers. If none exists, the level of risk associated with an important purchase situation is expected to increase. Second, if buyer–seller relationships and channels of communication do already exist, the relative strength and depth of these relationships and channels have the ability to influence perceived levels of purchase related risk.

The Risk Continuum

As illustrated in Figure 4B, we propose that the levels of risk associated with organizational purchase situations range on a continuum from low to high. Based on our review, it appears that different levels of risk affect the constructs group, participant, seller, and informational characteristics, as well as the constructs conflict/negotiation, decision rules, role stress, buyer–seller relationships, and communication networks. In general, *as the risk associated with an organizational purchase increases:*

- The buying group (center) will become larger and more complex. That is, more people will be involved in the purchase decision throughout the different stages of the purchase process and these people will be drawn from, or represent, a greater variety of departmental and/or organizational interests. Additionally, buying center participants will generally be of higher organizational status and authority, or, if not, the buying center will not have authority to make the final purchase decision. Instead, its function will be to: (1) gather and evaluate relevant information and (2) to make recommendations to one or more members of upper-level management.
- Participants in the purchase decision-making process will be more educated and possess greater levels of experience in their particular area of expertise. Also, as the importance associated with the purchase decision is high, participants will be motivated to expend greater effort and deliberate more carefully throughout the purchase process. This heightened level of motivation will be most keenly felt by those members of the buying center whose department or work associates will be directly impacted by the final purchase decision.
- Sellers who offer proven products and solutions will be favored. Product quality and after-sale service will be of the utmost importance. Price, while always important, will be considered only after product and service criteria have been fully met. In general, only when (after careful scrutiny) two or more sellers appear equally capable of satisfying purchase requirements, will price play a dominant role in the purchase decision.
- Information search will be active and a wide variety of information sources will be used to guide and support an important purchase decision. Buying center participants may rely more heavily on impersonal, commercial information sources (e.g., trade publications, sales literature) during the earlier stages of the decision process. However, as the procurement decision progresses, personal, noncommercial information sources (e.g., outside consultants, other organizations that have already made similar purchases) may become more important.
- Within the buying firm, conflict between buying center participants will increase because (1) more departments (through their representatives) will be involved in the purchase decision—a greater diversity in departmental per-

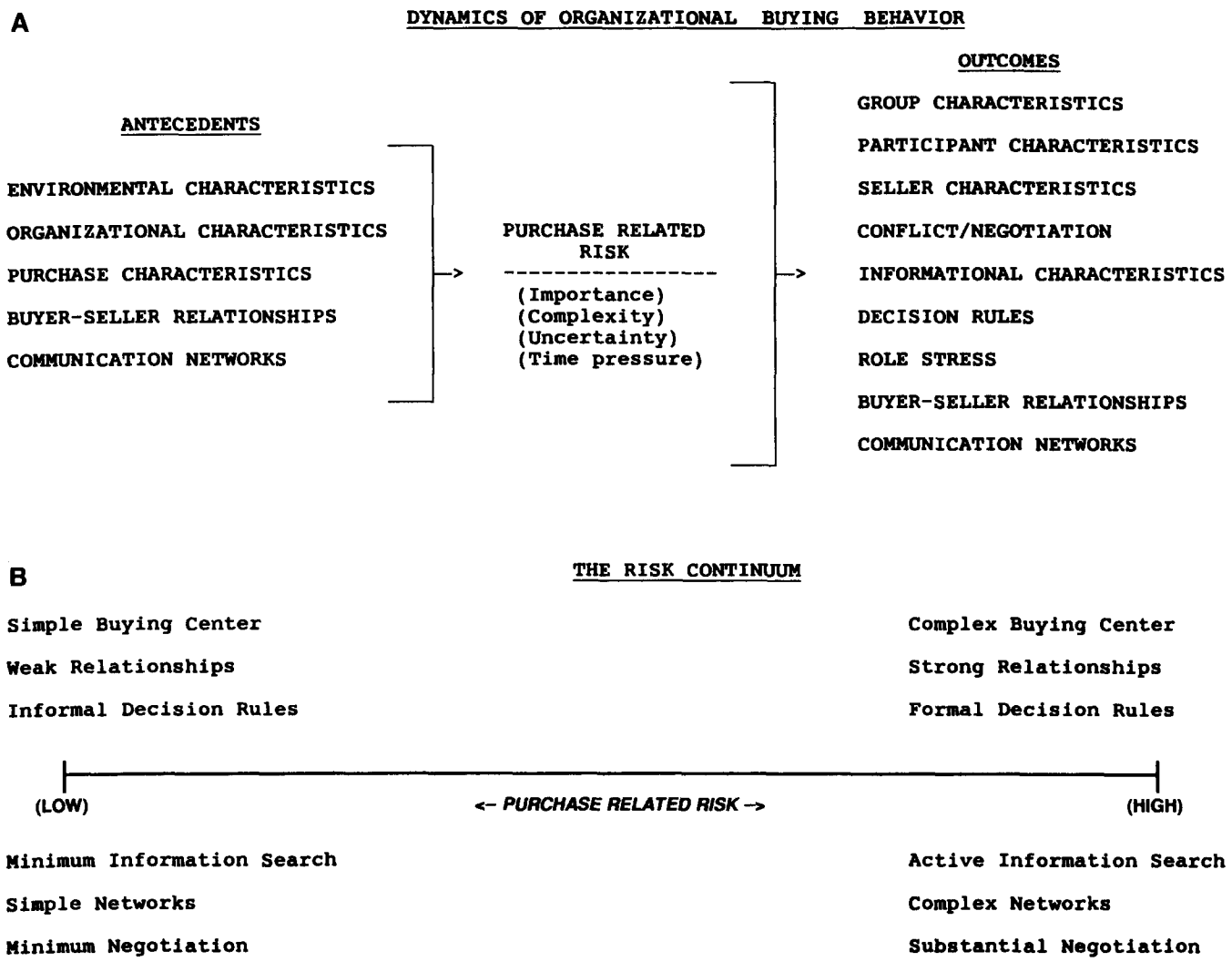


Figure 4. (A) Dynamics of organizational buying behavior; (B) the risk continuum.

spectives and motivations will lead to greater conflict, and (2) as the purchase outcome is important, buying center participants will be reluctant to make concessions without some form of reciprocal reward. Therefore, among buying center participants, a "bargaining" negotiation strategy will most likely be used. In contrast, however, the negotiation strategy most likely to be used between the buying and selling firms is a collaborative or problem-solving approach. In these cases, the buying firm's primary goal is to find the best solution to an important purchase problem and/or to reduce the risk associated with an uncertain purchase outcome. Consequently, rather than being confrontational or aggressive, negotiations between buyers and sellers focus on cooperation and information exchange in an attempt to discover the best solution to the purchase problem.

- The decision rules used in any given purchase situation are fundamentally firm specific. In general, however, most firms have some formalized purchase control mechanisms and decision-making guidelines, especially for recurring purchase situations. However, in new task purchase situations (where outcome uncertainty is increased) it may be difficult or inappropriate to use established procedures and guidelines. Thus, buyers may be forced to use a "decide-as-you-go" approach to the purchase decision process.
- Role stress will increase primarily due to two factors. First, as the size and complexity of the buying center increases, greater conflict among participants with differing perspectives and motivations is inevitable. And, second, in a highly visible (important) purchase where the outcome is uncertain, the chances of making a wrong decision and the

associated consequences of a wrong decision increase participant stress.

- Interfirm (buyer–seller) relationships and communication networks become increasingly important in higher risk purchase situations. Awarding the contract to a seller whose products and services have a proven intraorganizational track record helps reduce the perceived risk associated with an important purchase. Similarly, established networks of communication between multiple members of the buyer and seller firms facilitates information exchange and fosters an atmosphere of cooperation, further helping to reduce perceived risk.

Implications for Future Research

As far as we know, ours is the first attempt to accumulate and review the bulk of the past 25 years of research in organizational buying behavior. The 165 articles included in our analysis, although not exhaustive, represent most of the important work done in this area. Clearly, this research has contributed to and extended our knowledge of the organizational buying process. However, future contributions might be made in the following ways. First, our reference list could provide a rich base for a meta-analysis of some of the constructs included in our integrated model. Of particular interest would be a meta-analysis focusing on one or more of the four constructs that have most frequently been used in previous organizational buying behavior research. These constructs include: (1) purchase characteristics, (2) organizational characteristics, (3) group characteristics, and (4) participant(s) characteristics. Such an analysis could help to further organize and synthesize the contributions made by various works.

Second, additional empirical research on constructs that have been used less frequently in the study of organizational buying behavior could contribute to our overall understanding. In some cases, work already done in related fields could be helpful. For example, role stress has been examined by both psychology and organizational behavior researchers. Similarly, management researchers have studied the organization's external environment and its impact on the firm. Lastly, as previously discussed, organizational buying behavior researchers have rarely included the communication network construct in their work. However, our understanding of communication networks and how they influence organizational buying behavior could be enhanced by examining the extant work in the channels of distribution, personal selling and sales management, and psychology literatures.

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